Rules and networking concepts

Companion to v4.0 / For workshop facilitators / 20 October 2026

THE CORE LEARNING PROCESS

Destination IP > Network > Local routing table > Next hop

1 The board: nodes, links and topology

PCs and routers are nodes; their connections are links. Four connected routers form a ring topology. The sequence actually followed by a packet is its path. The topology stays fixed, but changing a route can change the path to a destination network.

2 Orders: destination IP and network

For example, 192.168.4.20 matches 192.168.4.0/24 under the common /24 mask. The order retains its destination IP while players look up the route again at each router. The next hop may change; the final receiver does not.

3 Starting at R1: the default gateway

Player PCs belong to 192.168.1.0/24. Their gateway is R1's LAN interface, 192.168.1.1. Traffic for B, C and D goes to this gateway first. R1 is a device name, 192.168.1.1 is an interface IP, and the player PC has its own IP.

4 Shared tables: hop-by-hop forwarding

Entries show a destination network, a common mask and a next-hop interface IP. R1 decides whom to forward to; it does not decide every later step for R2. At R2, R2's table controls forwarding. A directly connected local destination is delivered without selecting another neighbouring router.

Follow one order for network D

Source: 192.168.1.10/24 / Destination: 192.168.4.20/24

Initially, R1 routes D traffic through R2

Router	Matched network	Next hop
R1	192.168.4.0/24	R2 / 10.0.12.2
R2	192.168.4.0/24	R3 / 10.0.23.2
R3	192.168.4.0/24	R4 / 10.0.34.2
R4	192.168.4.0/24	Local delivery

Without congestion, three router-to-router links take three forwarding phases. After R1's D entry switches to R4 (10.0.14.2), the packet crosses only R1-R4. Entries at R2, R3 and R4 do not change.

Why is the next-hop IP different?

10.0.12.2 is R2's interface on the R1-R2 link: the next stop. 192.168.4.20 is the final host. Each inter-router link has its own /30 subnet; its two interfaces belong to that same link subnet.

Link and subnet	One interface	Other interface
R1-R2 / 10.0.12.0/30	R1 10.0.12.1	R2 10.0.12.2
R2-R3 / 10.0.23.0/30	R2 10.0.23.1	R3 10.0.23.2
R3-R4 / 10.0.34.0/30	R3 10.0.34.1	R4 10.0.34.2
R1-R4 / 10.0.14.0/30	R1 10.0.14.1	R4 10.0.14.2

Use these addresses for explanation or simulation; players need not calculate /30 subnets. Game tables show only the four LAN destinations and omit other connected-link and local-interface routes present on a real device.

What the flip card and queues teach

Manual static routes / Shared effects / Limited capacity and waiting

1 Manually replacing a static route

A player acts as an administrator, switching R1's next hop for 192.168.4.0/24 between 10.0.12.2 and 10.0.14.2. This replaces one static route: only one option is active. It does not install two equal-cost routes, and it is not automatic route selection by a dynamic routing protocol.

2 Shared changes affect matching traffic

During forwarding, all D packets still at R1 use its final entry. B and C packets are unaffected by this change. A D packet already at R2 uses R2's table and continues to R3. A shared change neither recalls packets nor changes their destination IP.

3 Limited capacity creates queues

Each link allows at most two packets per round; excess packets wait. Shared capacity illustrates contention and queuing delay. Waiting packets keep their place; rerouted packets join a new queue tail. Waiting does not mean packet loss, and two packets per round is not an actual bandwidth in bit/s.

THREE SIMPLIFICATIONS TO EXPLAIN

1. Rounds are uniform time units. One link per round controls game pace; processing, propagation, transmission and queuing delays are not calculated separately. Shared two-way capacity is not a model of a real full-duplex link.
2. Rechecking queued packets and moving them to another queue is a game rule. Real output queues do not necessarily reroute packets after a routing update.
3. Arrival at the destination router includes immediate host delivery. Game moves therefore do not directly count all real routing operations or TTL decrements.

Coverage and learning checks

Teach the core clearly; use instruction or Packet Tracer for the remaining topics

What learners practise

Identify a /24 destination network; use the default gateway; look up the destination IP at the current router; distinguish the next hop from the final destination; manually change a static route; observe shared routing effects and link queues.

What this game does not implement

Binary IPv4 representation, /26 subnet calculations, direct same-subnet communication, ARP, MAC and frame encapsulation, TTL, longest-prefix matching, ICMP replies, OSI/TCP/IP layer mapping, VLANs, SVIs and ACLs. IP-address cards alone do not constitute coverage of these topics.

An order checks one-way delivery, not a complete ping. Packet Tracer ping tests also require correct return routes, interface settings and host gateways. The shared tables include routes to A as a reference for return configuration, but the game does not move reply packets.

Three questions to ask after play

1. For 192.168.4.20, whom does R1 initially forward to?
Answer: R2, using next-hop interface IP 10.0.12.2.
2. Does switching R1's D route to R4 redirect a packet already at R2?
Answer: No. It still uses R2's table and goes to R3.
3. Does changing the route replace the destination IP with the next-hop IP?
Answer: No. Without NAT here, the destination remains 192.168.4.20.

Basis: the revised knowledge outline in 6108proj/ee6108.docx, with earlier outlines and the experiment list used to identify exclusions. Game reference: the v4 shared-routing-table rulebook. Use both PDFs together.